

Bias as a feature for generative AI detection

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Abstract

As large language models increasingly generate fluent text at scale, reliable human-AI authorship detection has become both important and difficult. This paper investigates whether AI-text detection can benefit from detecting bias-related features as a complementary signal beyond stylometry. In this paper, three logistic-regression classifiers are described: a baseline stylometric model using low-dimensional surface features, a bias-oriented model using lexicon-based and affective features and a merged model. Evaluation is conducted on the PAN 2025 Voight-Kampff Subtask 1 dataset and on a custom migration-focused dataset designed to control topic and genre. On PAN, the baseline model achieves 95.2 % accuracy, substantially outperforming the bias-oriented model (84.3 %), while the merged model yields the best overall performance (96.2 %). On the migration dataset, the baseline reaches 100 % accuracy under controlled conditions; the bias-oriented model attains 96.2 %. Error analysis shows that stylometric and bias features fail in systematically different ways, supporting the use of combined feature views for more robust detection.

1 Introduction

Ever since large language models (LLMs) have learned to produce fluent, human-like text at scale, there has been growing interest in human-AI authorship detection. Current approaches tend to fall into different broad families: methods that exploit properties of an LLM’s probability structure (e.g., DetectGPT; Mitchell et al. (2023)), supervised classifiers that learn to separate human and machine text from labeled examples (e.g., transformer-based models such as RoBERTa-BiLSTM; Bafna et al. (2024)) and statistical or stylometric methods that detect regularities in punctuation, function-word use, lexical diversity and related distributional cues (e.g., GLTR; Gehrmann et al. (2019)).

However, reliable detection has proven increasingly difficult: modern generators produce highly fluent and coherent text that can be hard to distinguish from human writing and simple paraphrasing can significantly reduce detector accuracy. Additionally, many detection models exhibit brittleness when applied to text from unseen generators or across domains with different characteristics, highlighting robustness challenges (Fu et al., 2025).

This paper asks **whether AI-text detection can benefit from an additional, complementary perspective: bias-related tendencies in content-level and rhetorical choices**. Studies show that large language models exhibit systematic social biases in their writing, often reproducing and amplifying stereotypes present in their training data regarding gender, racial and ethnic background, religion, national or geographic origin, age, disability or sexual orientation (Gallegos et al., 2024). LLMs additionally show content and style bias (Bang et al., 2024): they use generic language, hedge their statements, use less polarized vocabulary and remain more unsentimental and polite than human writers.

To test **how such bias features can add coverage to an AI-detection model**, I compare three lightweight linear classifiers trained with logistic regression: a baseline stylometric model, a bias-oriented model and a merged model that concatenates both feature sets. The models are evaluated on the PAN 2025 Voight-Kampff Subtask 1 benchmark (Greiner-Petter et al., 2025) and on an additional migration-focused dataset constructed to control topic and genre more tightly.

1.1 Related works

Early detection systems combined simple statistical cues with language-model likelihood information to help users spot generated text; GLTR, for instance, visualizes token likelihood patterns that differ across sampling schemes (Gehrmann et al., 2019). Work on neural fake-news generation and

defense (e.g., Grover, Zellers et al. (2020)) demonstrated that strong generators often enable strong discriminators, but also highlighted a detection-generation arms race.

More recently, zero-shot approaches such as DetectGPT leverage properties of a model’s probability landscape (via perturbations and curvature) to detect machine text without supervised training data (Mitchell et al., 2023). In parallel, many studies explore supervised transformer-based detectors; while effective in-domain, they may degrade under generator changes or genre/topic shifts (Bafna et al., 2024).

A growing literature examines how LLM outputs differ from human text in sentiment, toxicity and other socially salient dimensions. Prior work has documented stylistic moderation effects such as reduced sentiment intensity and hedging (Bang et al., 2024), as well as shifts toward more neutral sentiment in LLM-mediated rewriting in some settings (Wang et al., 2025). Work on toxicity evaluation further suggests that model outputs and their harmfulness profiles depend strongly on prompting and context (Luong et al., 2024). These findings motivate exploring bias-related signals as detection features, while also cautioning that such signals can be entangled with topic and genre.

2 Methods

To extract and systematically test features on AI- and human-generated text, three linear classifiers were trained using logistic regression: a baseline stylometric model, a bias-oriented content-level model and a merged model.

2.1 Baseline model

The baseline model was constructed from hand-crafted, low-dimensional stylometric features designed to capture **surface-level linguistic regularities**, including sentence and word length, stop-word rate, punctuation frequency, character-class ratios (uppercase, title case, digits, whitespace), type-token ratio, UPOS tag frequencies, 5-gram repetition and self-similarity (for details, see Table 6 in Appendix).

Each feature was developed and its relevance was tested on the training data from Subtask 1 of the PAN 2025 Voight-Kampff challenge. No information from the validation split was used for feature development or model fitting.

2.1.1 Features discarded from the baseline model

During early development, character- and word-level n-gram features using TF-IDF representations were also explored. These features were excluded from the baseline model because the resulting feature space was extremely high-dimensional relative to the size of the dataset, yielding a sparse representation that showed substantially higher performance on the training data than on the testing data. This discrepancy suggested overfitting, with the classifier capturing surface-level artifacts specific to the training data.

2.2 Bias-oriented model

A second feature set was designed to capture **potential tendencies related to content-level, stylistic and rhetorical biases** that differ between human- and AI-generated texts, including sentence-level sentiment, lexicon-based positive and negative tone, hedging and modality markers, subjectivity markers, profanity, categorical language and assertiveness (for details, see Table 7 in Appendix).

The bias features were more exploratory than the baseline features, with some overlap between them. For example, the emotional tone feature conceptually overlaps with sentiment analysis but relies on transparent lexical counts rather than a pre-trained model. Similarly, the subjectivity feature is correlated with sentiment analysis and partially overlaps with the hedging lexicon, but it more broadly captures evaluative and opinion-based language.

As with the baseline model, each feature was developed and its relevance evaluated on the training data from the Subtask 1 of the PAN 2025 Voight-Kampff challenge. As a result, several bias-related features were explored but ultimately excluded.

2.2.1 Features discarded from the bias-oriented model

- A lexicon of emojis and emoticons was explored, but in the PAN dataset, no emojis were observed, making the feature effectively constant.
- Markers for politeness and impoliteness were extremely sparse in the PAN dataset, leading to unstable estimates when using a frequency feature.
- A pre-trained political stance classifier led to sparse and unstable outputs across the het-

erogeneous genres in the PAN dataset and showed minimal class separation.

- A measured ratio of group markers “we” to “they” was unstable due to a very small number of numerators and denominators found in the texts.
- Semantic or embedding-based bias representations (e.g. Word2Vec similarity) were considered but rejected due to the short length of each text.

For consistency, these features were also excluded from subsequent experiments on the migration-specific dataset.

2.3 Merged model

The merged model combined the baseline stylistic features and bias-related features. For each document, the **baseline and bias feature vectors were concatenated** into a single feature vector. Across documents, these formed the final feature matrix. Feature extraction, scaling and classification followed the same procedure as in the individual models.

3 Data

3.1 PAN 2025 Voight-Kampff dataset

The primary dataset was obtained from **Subtask 1 of the PAN 2025 Voight-Kampff challenge, which poses a binary AI-detection task (human-authored vs machine-authored)**. The dataset contains texts from multiple genres authored by humans and generated by large language models (such as DeepSeek, Gemini, GPT and Llama) and was provided with a predefined training-validation split of 87 % / 13 %. The validation texts were used as the held-out test set in this project. The detailed distribution is reported in Appendix Tables 8 and 9.

Of the 27,296 total texts, the class distribution was moderately imbalanced toward AI, with approximately 62 % AI-generated and 38 % human texts. This distribution was broadly similar across the training and validation splits. Accordingly, a majority-class baseline that always predicts the most frequent label (AI) achieves approximately 64 % accuracy on the validation split, so model performance should be interpreted accordingly.

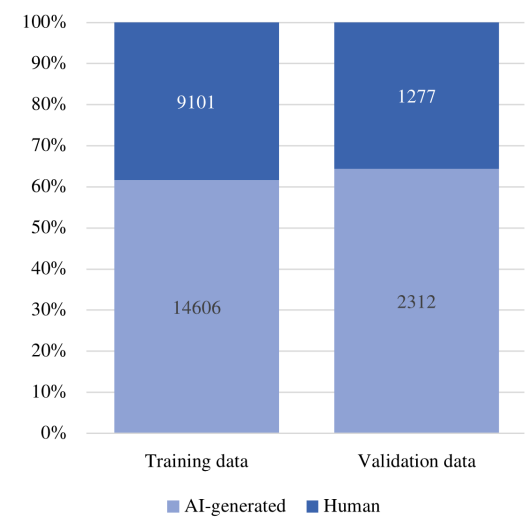


Figure 1: Data distribution in the PAN dataset

3.2 Custom migration-focused dataset

To evaluate model behavior under stricter topical and genre control, an additional dataset focused on migration was constructed. This topic was chosen in the hope that political controversy might lead to higher bias scores in the data.

3.2.1 Human corpus collection

Human-written texts were drawn from the SFU Opinion and Comments Corpus (Kolhatkar et al., 2020). This large corpus consists of editorials, opinion columns and op-eds published in The Globe and Mail between January 2012 and December 2016. This source was chosen for several reasons: the texts are explicitly opinionated and argumentative; they were written by journalists before the widespread use of large language models; and they originate from a single publication, reducing stylistic variance.

As a first step, the 10,366 articles from the corpus were searched and only the 466 articles containing at least two migration-related terms from a predefined lexicon (e.g., immigration, migrant, refugee, asylum, border, deportation, undocumented, visa, integration) were retained. To further filter migration-focused texts, a zero-shot topic classification step was applied using a BART-based natural language inference model. Articles were classified into four topical categories, one of which was migration-specific. Based on content relevance and argumentative focus, a corpus of **300 migration-focused opinion texts by human writers** was built.

3.2.2 AI-generated corpus collection

To create comparable AI-generated texts, 100 opinion-style articles each were compiled using OpenAI’s GPT model, Google Gemini and DeepSeek.

To match the genre and communicative intent of the human texts, prompts explicitly requested newspaper editorials or opinion pieces written for a general audience and focusing on migration. To reduce prompt-specific artifacts, a small set of variants was used (e.g., “Write a newspaper opinion piece about immigration”, “Write an editorial on immigration for a general newspaper audience”) and prompts were randomly sampled across generations. Text length was also controlled explicitly, matching the character-based length distribution of human texts.

The complete dataset was then split into training and test sets using the same 87 % / 13 % split as the PAN dataset, applying stratification to preserve balance across human vs. each AI model and text length. **The final dataset contained 600 texts (300 human, 300 AI): 522 for training and 78 for testing.**

4 Results

4.1 Baseline model performance

4.1.1 PAN dataset

On the PAN 2025 Voight-Kampff Subtask 1 validation set, the baseline model achieved an accuracy of 95.2 % with a recall above 94 % for both human and AI.

Coefficient analysis of the trained logistic regression model revealed that the features with the strongest positive weights toward the AI class include higher pronoun and determiner frequencies, higher type-token ratio, longer average word length and higher repetition and self-similarity scores. In contrast, human-written texts are characterized by higher stop-word rates, greater sentence-length variability, higher rates of proper nouns and increased use of punctuation such as semicolons.

4.1.2 Migration dataset

When evaluated on the custom migration-focused dataset, the baseline model achieved perfect classification performance, correctly classifying all 78 test instances.

Given these unusually strong results, additional checks were conducted to rule out trivial explanations. First, all explicit length-based features

Class	Recall	Precision	F1-Score
Human	0.948	0.915	0.931
AI	0.954	0.972	0.963
Accuracy		0.952	

Table 1: Results of baseline model on PAN dataset

Class	Recall	Precision	F1-Score
Human	1.000	1.000	1.000
AI	1.000	1.000	1.000
Accuracy		1.000	

Table 2: Results of baseline model on migration-specific dataset

Class	Recall	Precision	F1-Score
Human	0.793	0.757	0.775
AI	0.869	0.891	0.880
Accuracy		0.843	

Table 3: Results of bias-oriented model on PAN dataset

Class	Recall	Precision	F1-Score
Human	0.973	0.947	0.960
AI	0.951	0.975	0.963
Accuracy		0.962	

Table 4: Results of bias-oriented model on migration-specific dataset

Class	Recall	Precision	F1-Score
Human	0.960	0.931	0.945
AI	0.963	0.978	0.970
Accuracy		0.962	

Table 5: Results of merged model on PAN dataset

were removed to test whether the model was exploiting length-related leakage; accuracy remained 100 %. Second, the model was trained using only GPT- and Gemini-generated text and evaluated on held-out DeepSeek-generated texts; accuracy again remained 100 %. These checks suggest the observed performance is unlikely to be explained by length as a confound, train-test contamination, or generator-specific memorization.

Coefficient analysis on the migration-specific dataset shows a broadly similar pattern of influential features as in the PAN dataset: AI predictions were associated with higher noun, pronoun and determiner frequencies, as well as increased word-length variability, while human texts were more strongly associated with proper nouns, sentence-length variability and casing-related features.

4.2 Bias-oriented model performance

4.2.1 PAN dataset

On the PAN 2025 Voight-Kampff Subtask 1 validation set, the bias-focused model achieved a moderate overall accuracy of 84.3 %, substantially lower than the baseline stylometric model but still clearly above the majority-class baseline. Performance was again higher for AI-generated texts than for human-written texts, with a recall of 0.891 for AI compared to 0.757 for human texts.

Coefficient analysis indicates that sentiment- and emotion-related features dominate the bias-oriented model’s decisions. Strong positive weights toward the AI class were observed for negative and positive emotional tone rates, as well as mean sentiment score. In contrast, human-written texts were more strongly associated with higher rates of identity-related terms (gender, race/ethnicity, nationality), profanity and assertive language.

Overall, the bias-oriented model captures differences in emotional polarity, evaluative language and identity references, but lacks the fine-grained stylistic sensitivity of the baseline model, resulting in substantially lower performance on the heterogeneous PAN dataset.

4.2.2 Migration dataset

When evaluated on the custom migration-focused dataset, the bias-oriented model achieved high classification performance, reaching an accuracy of 96.2 % on the held-out test set.

Feature-level analysis reveals a pattern consistent with the PAN results but with stronger and more concentrated weights. AI predictions were

again driven primarily by higher positive and negative emotional tone rates and higher mean sentiment scores. Human-written texts were more strongly associated with higher rates of identity-related terms, profanity and assertive language.

Compared to the PAN dataset, the bias features appear substantially more effective in the migration-specific setting, likely reflecting the topic’s inherently evaluative and polarized nature, which amplifies emotional and identity-related signals captured by the bias feature set.

4.3 Merged model performance

4.3.1 PAN dataset

The merged model, which combined all baseline stylometric features with the bias-oriented lexical and affective features into a single classifier reached an accuracy of 96.2 %, achieving the highest overall performance among all evaluated models.

Coefficient analysis shows that the merged model is dominated by baseline stylometric features, but that bias features contribute complementary signal. The strongest weights pushing predictions toward the AI class are primarily baseline features, such as higher pronoun and determiner frequencies, higher type-token ratio, longer average word length and increased repetition and self-similarity. However, several bias features also appear among the top contributors, most notably negative and positive emotional word rates and mean sentiment score.

Conversely, predictions toward the human class are again driven mainly by baseline features such as higher stop-word rates, greater sentence-length variability, increased use of proper nouns and specific punctuation patterns. Bias features such as profanity rate, polarity score, categorical language and nationality-related identity terms also contribute negatively, reinforcing human predictions.

4.3.2 Migration dataset

The merged model was not evaluated on the custom migration-specific dataset, since the baseline model already achieved perfect classification accuracy (100%). Under these conditions, adding bias features couldn’t possibly improve measurable performance or provide additional insight.

5 Discussion

5.1 Misclassifications

5.1.1 Overlapping misclassifications

Across all three models, 66 texts were consistently misclassified. A matrix overview of all misclassifications shared and specific to models is attached in Table 10 in the Appendix. An error analysis of these cases shows that the baseline and bias-oriented models are not failing for opposing or competing reasons. Instead, they fail for the same underlying reason and therefore do not “contradict each other” on these texts.

These instances are best characterized as globally ambiguous texts. They do not exhibit strong signals in either feature space: stylometric features show no pronounced deviation toward AI-like or human-like patterns; values largely cluster around the training mean. Bias-related features likewise lack extreme values that would clearly indicate human stance-taking or AI-like neutrality. As a result, neither model produces strong, confident evidence in favor of one class.

Crucially, these are not conflicted texts, where different feature groups point in opposite directions. Rather, they are cases of signal absence: both feature spaces are weakly informative. This explains why the merged model does not recover these errors. Feature stacking is effective only when complementary signals exist across feature sets. Consequently, in these cases, merging baseline and bias features does not improve classification performance, because the underlying ambiguity persists across all representations.

5.1.2 Baseline model misclassifications

All texts in the baseline-specific misclassifications exhibit unusually strong and internally consistent stylistic profiles, deviating sharply from typical examples of their true class across multiple features. In stylistic space, they more closely resemble the opposing class than their true label, making the error systematic rather than accidental.

Importantly, these errors are not associated with weak or noisy feature representations. On the contrary, they are among the most stylometrically distinctive texts relative to the training distribution. These findings indicate that the baseline model fails when stylometric cues are actively misleading rather than absent. Specifically, it struggled with AI-generated news texts, which presumably closely imitate the structural properties of human

journalism. Similarly, highly polished human texts that exhibit unusually regular structure and reduced idiosyncratic variation also posed a challenge.

5.1.3 Bias-oriented model misclassifications

Bias-oriented model-specific misclassifications exhibited a markedly different profile from baseline-specific errors. They typically showed low to moderate assertive rates across most texts, with limited variance (in particular for sentiment scores) and very low identity-term rates across all identity categories.

In contrast to baseline errors, these misclassifications do not arise from unusually marked feature patterns. Instead, they involve texts whose content-level signals remain close to the model’s expectations for both classes. The bias-oriented model fails primarily on content-neutral or weakly opinionated texts, where sentiment, identity and stance-related cues are sparse or absent. Rather than presenting conflicting evidence, these texts simply offer too little signal for reliable discrimination. This includes straight news reporting, neutral essays, factual summaries and fiction without a clear moral or ideological stance.

5.1.4 Merged model misclassifications

The merged model benefits from combining the two complementary perspectives of the baseline and bias-oriented models, which substantially reduces brittleness relative to the single-view approach. This is most clearly illustrated by a small but informative subset of 11 texts that were misclassified by both the baseline and bias-oriented models but correctly classified by the merged model. Notably, 8 of these 11 texts (73 %) are long-form news explainers characterized by structured, factual and neutral reporting.

5.2 Differences between the datasets

The difference in performance between the PAN dataset and my own migration-specific dataset is best explained by structural properties of the datasets rather than model behavior. The PAN dataset spans a wide range of topics, genres and writing conditions, including news, fiction, essays and multiple styles of human authorship. This diversity leads to substantial stylistic overlap between human and AI-generated texts. As a result, stylometric features such as sentence length, POS balance, lexical richness and punctuation regularity do not consistently separate the two classes. Even

highly regular writing can plausibly originate from either humans or LLMs, which limits separability and results in a performance ceiling of approximately 95 % for the baseline model.

In contrast, the migration-specific dataset is highly constrained. All texts address a single topic and belong to a single genre (opinion/editorial), with tightly controlled prompting conditions. This restriction concentrates the stylistic signal: human texts exhibit consistent author-specific variation, while AI-generated texts display uniform regularities associated with prompt-following and generation smoothing. Under these conditions, surface-level stylometric features become highly discriminative, allowing perfect separation, even when evaluated on an unseen generator.

Importantly, the migration-specific dataset is also small and was compiled with a limited set of prompts. Under such circumstances, a model can exploit stable editorial conventions and prompt-induced regularities, without true generalizability. This result is therefore explicitly in-domain and does not imply robustness across domains or genres. The observed performance should therefore be interpreted as an upper bound under idealized conditions, rather than evidence of real-world deployment readiness.

6 Conclusion

Given the strong standalone performance of the baseline model shown in this study, one might reasonably ask whether exploring bias features is necessary at all. At present, stylometric features remain highly effective, often outperforming content-based signals by a wide margin. However, this advantage is unlikely to persist indefinitely. As large language models continue to improve and as human writers increasingly adapt to AI-mediated writing norms, purely stylistic differences are likely to erode. Moreover, models can already be prompted to produce text that deliberately mimics human stylistic variation in order to evade form-based detection.

This study shows that stylometric and content-based bias features fail in systematically different ways. Baseline-only misclassifications arise when structural cues are misleading: texts are well-formed, stylistically regular and genre-appropriate, even though their content provides little discriminatory signal. In contrast, bias-oriented model-only misclassifications arise when content cues are

absent: texts are informational, neutral and low in stance or identity signaling, even though their structure is unremarkable and human-like.

This complementarity explains why the merged model substantially reduces bias-oriented model-specific errors. Stylometric features compensate precisely in cases where content-based signals collapse, while bias features provide coverage where stylistic cues are genre-confounded or misleading. Crucially, the merged model does not simply succeed by stacking more features. Instead, it recovers cases where neither stylistic deviation nor bias cues alone are sufficient, but where their interaction exposes inconsistencies between surface regularity and semantic intent. These structural-semantic mismatches are characteristic of advanced AI-generated text and are only detectable when both perspectives are evaluated jointly.

However, several limitations must be acknowledged. First, bias features are inherently correlated with topic and genre. For example, news articles about technical infrastructure must be expected to exhibit lower bias and stance signals than opinion pieces on contentious social issues, regardless of whether they are written by humans or AI. As a result, content-based features inevitably reflect dataset composition in addition to differences between human and AI authorship.

Second, many features commonly associated with AI-generated text, such as politeness, positivity, or certain punctuation patterns, are also part of legitimate human writing styles. Humans may naturally write in a highly regular, polished manner or adopt stylistic conventions that resemble those of language models. This makes false positives, particularly for human texts, a persistent risk and highlights the importance of maintaining high human recall in any detection system.

7 Tools

All experiments were conducted using Python. Code and results are published on github: <https://github.com/erscheckend/ai-detection-with-bias>

Text preprocessing and linguistic annotation were performed with **spaCy**, including tokenization, sentence segmentation, part-of-speech tagging and stop-word identification. Sentiment analysis was implemented using the **VADER** sentiment analyzer.

Feature extraction and statistical analysis relied

on **NumPy** and **pandas**, while model training and evaluation were carried out using **scikit-learn**, including logistic regression, feature scaling and performance metrics. Zero-shot topic classification for dataset construction was performed with a **BART-based natural language inference model** (facebook/bart-large-mnli).

Data visualization was implemented using **matplotlib**. All experiments were run on a local machine without GPU acceleration.

The text generation for the AI-generated texts of the migration-specific dataset was done using the models GPT 4o-mini (OpenAI), gemini-flash-lite-latest (Google) and DeepSeek (DeepSeek, accessed via <https://api.deepseek.com>). The code for the text generation is also visible in the github repository.

8 Declaration of independence

I hereby declare that I have written the present paper independently, without outside help and without unauthorized assistance and that I have identified all literal and paraphrased quotations from other works as such.

ChatGPT (OpenAI, <https://chatgpt.com>) was used as follows:

- Searching relevant research and ideas about features to explore,
- Debugging code blocks,
- Checking the rough structure and argumentation for the paper,
- Linguistic revision of the paper.

In addition, the generative AI models GPT 4o, DeepSeek and Gemini were used in text generation for one dataset, but not applied in any further capacity for the paper or code.

Despite the use of these tools, I declare that this work is essentially my own achievement.

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A Appendix

Feature	Information captured
Sentence- and word-length	Sentence lengths as number of tokens per sentence, word lengths as number of characters per token.
Rate of stop-words	Stop-word count over total non-punctuation tokens.
Frequency of punctuation marks	Occurrence of each selected mark (, . ; : ? ! ' " () - ...) in raw text, normalized by total characters and scaled to 1,000 characters.
Uppercase, title case, digits, whitespaces	Ratio of uppercase, digits and whitespace characters per total characters. Tokens with title case per number of whitespace-split tokens.
Type-token ratio	Number of unique types per number of tokens.
Frequencies of Universal POS tags	Relative frequencies for the UPOS tags NOUN, VERB, ADJ, ADV, PRON, ADP, DET, CCONJ, SCONJ, NUM, AUX, INTJ, PART, PROPN.
Repeated 5-gram sequence	Proportion of tokens that belong to any 5-gram occurring more than once, computed over total tokens.
Jaccard similarity	Similarity between the 200 most frequent unigrams in the first and second halves of each text, computed over total tokens.

Table 6: Features of the baseline model

Feature	Information captured
Sentiment	Mean and variance of VADER compound sentiment scores across sentences.
Emotional tone	Rate of positive and negative terms from two lexicons, along with a polarity score derived from their difference.
Hedging and modality	Rate of terms from a lexicon of modal verbs (e.g. may, might), softening adverbs (e.g. perhaps, presumably), epistemic verbs (e.g. seem, suppose) and multi-word phrases (e.g. in my opinion, there is a chance).
Subjectivity	Rate of terms from a lexicon of personal perspective indicators (e.g. first-person opinion markers).
Profanity	Rate of terms from a lexicon of toxic, profane and insulting words.
Categorical statements	Rate of terms from a lexicon of categorical statements as expressed through absolute or extreme expressions (e.g. always, never, everyone).
Assertiveness	Rate of terms from a lexicon of forceful argumentative or strongly assertive terms (e.g. prove, must, it is certain).

Table 7: Features of the bias-oriented model

	Essays	Fiction	News	Total per model
deepseek-r1-distill-qwen-32b	173	728	0	901
falcon3-10b-instruct	175	704	0	879
gemini-1.5-pro	395	677	0	1072
gemini-2.0-flash	377	702	0	1079
gemini-pro	0	0	276	276
gemini-pro-paraphrase	0	0	265	265
gpt-3.5-turbo	398	706	270	1374
gpt-4-turbo	0	0	272	272
gpt-4-turbo-paraphrase	0	0	276	276
gpt-4.5-preview	150	128	0	278
gpt-4o	374	703	259	1336
gpt-4o-mini	390	704	264	1358
llama-2-70b-chat	0	0	269	269
llama-2-7b-chat	0	0	262	262
llama-3.1-8b-instruct	372	691	0	1063
llama-3.3-70b-instruct	167	238	0	405
ministral-8b-instruct-2410	391	709	0	1100
mistral-7b-instruct-v0.2	0	0	266	266
mixtral-8x7b-instruct-v0.1	0	0	264	264
o3-mini	368	707	0	1075
qwen1.5-72b-chat-8bit	0	0	271	271
text-bison-002	0	0	265	265
Human	931	7300	870	9101
Total per genre	4661	14697	4349	23707

Table 8: Distribution of texts across genres and sources (training split).

	Essays	Fiction	News	Total per model
deepseek-r1-distill-qwen-32b	29	77	0	106
falcon3-10b-instruct	25	88	0	113
gemini-1.5-pro	48	97	0	145
gemini-2.0-flash	56	83	0	139
gemini-pro	0	0	59	59
gemini-pro-paraphrase	0	0	70	70
gpt-3.5-turbo	53	82	65	200
gpt-4-turbo	0	0	63	63
gpt-4-turbo-paraphrase	0	0	59	59
gpt-4.5-preview	25	20	0	45
gpt-4o	54	79	76	209
gpt-4o-mini	55	88	71	214
llama-2-70b-chat	0	0	65	65
llama-2-7b-chat	0	0	72	72
llama-3.1-8b-instruct	60	98	0	158
llama-3.3-70b-instruct	25	29	0	54
ministral-8b-instruct-2410	51	88	0	139
mistral-7b-instruct-v0.2	0	0	68	68
mixtral-8x7b-instruct-v0.1	0	0	70	70
o3-mini	52	80	0	132
qwen1.5-72b-chat-8bit	0	0	63	63
text-bison-002	0	0	69	69
Human	132	928	217	1277
Total per genre	665	1837	1087	3589

Table 9: Distribution of texts across genres and sources (validation split).

	Baseline model	Bias model	Merged model	Shared (all)	Total per model
Baseline model	45	11	50	66	172
Bias model	11	472	13	66	562
Merged model	50	13	9	66	138

Table 10: Misclassification matrix